Name: Class: Date:

Activity 1 · Hands-on · 🏫 In class · ⏱ 40 minutes

✂️ Paper Plasmid: Make Recombinant DNA

You will learn: A restriction enzyme cuts DNA at a specific site leaving sticky ends; ligase joins a gene into a plasmid — how human insulin and Bt cotton are made.

YOU NEED

- Paper strips with printed DNA sequences (one "plasmid" circle and one "human insulin gene" strip)
- Scissors
- Tape (DNA ligase!)
- Highlighter

⚠️ **Safety:** Use scissors carefully.

STEPS

1. Write a plasmid ring: ...ATCGAATTCGCT... and a gene strip: ...GAATTC[insulin gene]GAATTC....
2. Highlight the EcoRI site GAATTC on both.
3. "Cut" with scissors between G and A on both strands, leaving overhanging sticky ends.
4. Fit the gene's sticky ends into the opened plasmid.
5. Join with tape — this is DNA ligase. You now have recombinant DNA.

OBSERVATION TABLE

Step	Molecular tool used	What happened

QUESTIONS

1. Why must the same restriction enzyme cut both the plasmid and the gene?

2. What are sticky ends?

3. Which organism was used to make human insulin chains?

WHAT IS HAPPENING? — THE SCIENCE

Restriction enzymes like EcoRI recognise palindromic sites (GAATTC) and cut between G and A, giving sticky ends. The same enzyme cuts the vector and the gene, so their ends match. Ligase seals them. The plasmid is then put into E. coli, which makes the protein — as Eli Lilly did for human insulin.

Key terms: restriction enzyme · sticky ends · DNA ligase · plasmid · recombinant DNA

Activity 2 · Hands-on · 🏠 At home · ⏱ 30 minutes

🥬 Red Cabbage Test: Why Bt Toxin Kills Only Insects

You will learn: Bt protoxin becomes active only in the alkaline gut of insects; our stomach is acidic.

YOU NEED

- A quarter of a red (purple) cabbage
- Hot water, a strainer
- Small clear glasses
- Lemon juice, vinegar, baking soda solution, soap water, plain water

⚠️ **Safety:** Handle hot water with an adult. Do not taste the test liquids.

STEPS

1. Chop the cabbage, soak in hot water for 10 minutes and strain — this purple liquid is a pH indicator.
2. Pour the indicator into 5 glasses.
3. Add lemon juice, vinegar, plain water, baking soda solution and soap water — one to each.
4. Note the colour: red/pink = acidic, purple = neutral, green/yellow = alkaline.
5. Decide which liquids are like the insect gut and which are like the human stomach.

OBSERVATION TABLE

Liquid	Colour	Acidic / neutral / alkaline	Like which gut?

QUESTIONS

1. Which liquids were alkaline?

2. Why is Bt protoxin harmless to the plant that makes it?

3. Name the genes that protect cotton from bollworms.

WHAT IS HAPPENING? — THE SCIENCE

Red cabbage contains anthocyanin, which changes colour with pH. The Bt Cry protein is made as an inactive protoxin; the alkaline pH of the insect midgut converts it into the active toxin, which binds the gut lining and makes pores. The human stomach is acidic, so the toxin is not activated in us.

Key terms: **Bt toxin** **protoxin** **Cry protein** **alkaline pH**

Digital activity

💻 Biotech in Real Life

> Tap an application, then tap its field.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch10/2**

AGRICULTURE · MEDICINE · DIAGNOSIS · ETHICS & LAW

Agriculture	Medicine	Diagnosis	Ethics & law

Bt cotton · Golden rice · RNAi against nematodes · Human insulin from E. coli · Gene therapy for ADA deficiency · α-1-antitrypsin from transgenic animals · PCR · ELISA · DNA probe with autoradiography · GEAC approval · Biopiracy · Basmati patent

> **You will learn:** Biotechnology helps farming, medicine and diagnosis — and needs rules to keep it safe and fair.

THINK & WRITE

1. Which diagnostic method detects very low amounts of a pathogen's DNA?

2. What is biopiracy?